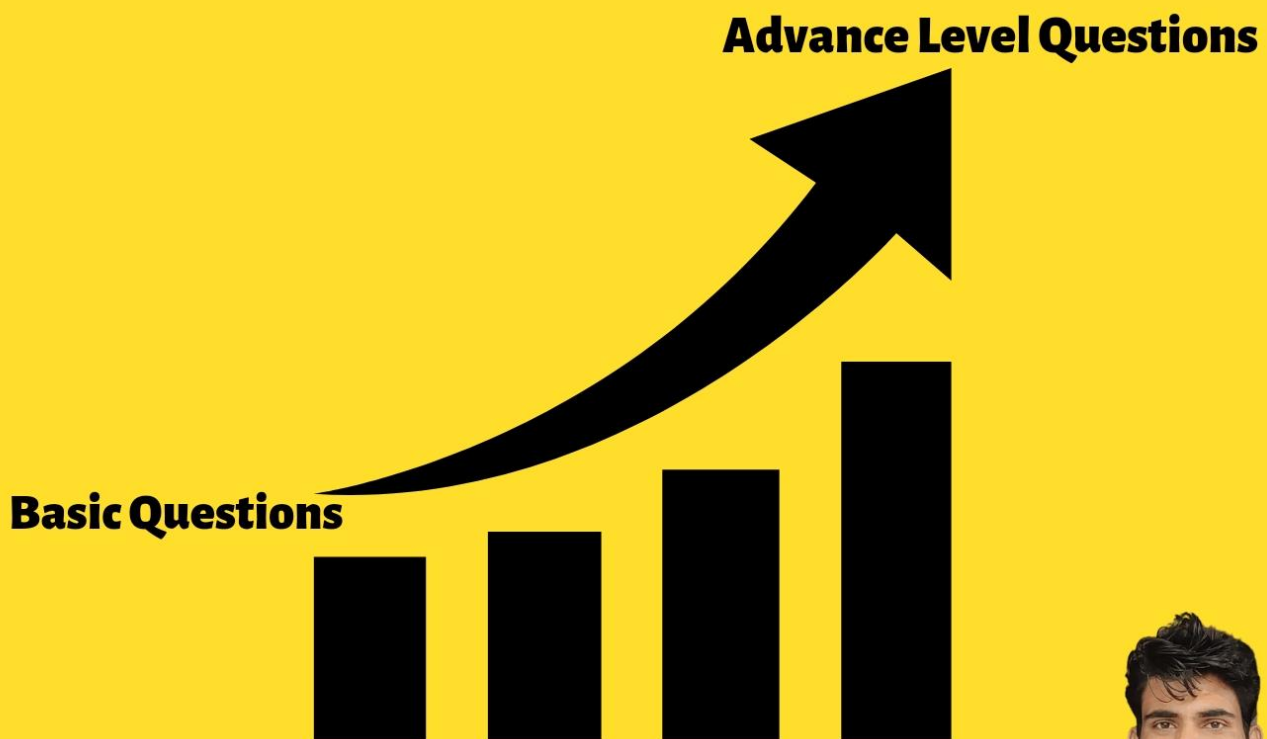


MATHS

PIPE & CISTERN

ALL TYPE WISE QUESTIONS



Maths with Gajanand



Pipe And Cistern

- 1] Tap A can fill a tank in 6 hours, Tap B can fill the same tank in 8 hours and Tap C can empty the same tank in 4 hours. If all three taps A, B and C are opened together, then how much time (in hours) will be taken to fill the tank?

Soln:-

$$\begin{array}{rcl}
 \text{eff.} & & \\
 4 \leftarrow A \rightarrow & 6 \text{ hr} & \\
 3 \leftarrow B \rightarrow & 8 \text{ hr.} & \\
 -6 \leftarrow C \rightarrow & 4 \text{ hrs} & \\
 \hline
 4+3-6 \leftarrow A+B+C & & \\
 1 \leftarrow A+B+C & &
 \end{array}
 \begin{array}{l}
 > 24 \\
 \text{Time} = \frac{24}{1} \text{ hrs.} \\
 = \boxed{24} \text{ Ans}
 \end{array}$$

- 2] A Pipe can fill a tank in x hours and another can empty it in y hours. In how many hours they together fill it if ($y > x$)

- (a) $(x-y)$ hrs (b) $(y-x)$ hrs (c) $\frac{xy}{x-y}$ (d) $\frac{xy}{y-x}$

Soln:- [Basic]

$$\begin{array}{l}
 \text{Ist :- In one hour} \rightarrow \frac{1}{x} \\
 \text{IInd :- In one hour} \rightarrow \frac{1}{y} \\
 \text{Both in 1 hour} \rightarrow \frac{1}{x} - \frac{1}{y} \\
 \Rightarrow \frac{y-x}{xy} \\
 \text{Time} = \frac{xy}{y-x} \text{ Ans}
 \end{array}$$

[m-II]

$$\begin{array}{rcl}
 y \leftarrow A \rightarrow & x & \\
 -x \leftarrow B \rightarrow & y & \\
 & > xy & \\
 \text{Time (A+B)} = \frac{xy}{y-x} \\
 \underline{\underline{\text{Ans}}}
 \end{array}$$

[3] A water tap fills a tub in 'p' hours and a sink at the bottom empties it in 'q' hours. If 'p < q' and both tap and sink are opened the tank is filled in 'r' hours; then the relation between p, q, r is:

(a) $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ (b) $\frac{1}{r} = \frac{1}{p} - \frac{1}{q}$ (c) $r = p + q$ (d) $r = p - q$

Soln :-

$$r = \frac{pq}{q-p}$$

[Formula:- $\frac{xy}{y-x}$]

$$\frac{1}{r} = \frac{q-p}{pq}$$

$$\boxed{\frac{1}{r} = \frac{1}{p} - \frac{1}{q}}$$

[4] Tap A can fill a tank in 8 hours, tap B can empty it in 10 hours. In how many hours they together fill it.

Soln :- m-I

eff.

$$5 \leftarrow A \rightarrow 8 \text{ hr.}$$

$$-4 \leftarrow B \rightarrow 10 \text{ hr}$$

$$> 40$$

$$5-4 \leftarrow A+B$$

$$= 1 \leftarrow A+B$$

$$A+B \rightarrow \frac{40}{1} = \boxed{40 \text{ hours}} \underline{\underline{\text{Ans}}}$$

m-II Trick

$$A \rightarrow 8$$

$$-B \rightarrow 10$$

$$A+B = \frac{10 \times 8}{10-8} \left[\frac{xy}{y-x} \right]$$

$$= \boxed{40 \text{ hours}} \underline{\underline{\text{Ans}}}$$

- [5] Pipes A, B and C can fill an empty tank in $\frac{30}{7}$ hours, if all the three pipes are opened simultaneously. A and B are filling pipes and C is an emptying pipe. Pipe A can fill tank in 15 hours and pipe C can empty it in 12 hours. In how long (in hours) can pipe B alone fill the empty tank?

Soln:-

eff.			
4	← A →	15 hr.	
-5	← C →	12 hr	
14	← A+B+C →	$\frac{30}{7}$ hr	
	↓ ↓		
4		-5	

LCM(15, 12, 30)
60

B = 15 B → $\frac{60}{15} = \boxed{4 \text{ hr}}$ Ans

- [6] Pipes A and B can fill a tank in 12 minutes and 15 minutes, respectively. The tank when full can be emptied by pipe C in x minutes. When all the three pipes are opened simultaneously, the tank is full in 10 minutes. The value of x is:

Soln:-

5	← A →	12 min.	
4	← B →	15 min.	
6	← A+B+C →	10 min.	
	↓ ↓ ↓		
5	4	-3	

60

C → $\frac{60}{3}$
= $\boxed{20 \text{ minutes}}$ Ans

- 7 Three taps A, B and C together can fill an empty cistern in 10 minutes. The tap A alone can fill it in 30 minutes and the tap B alone in 40 minutes. How long will the tap C alone take to fill it?

Soln:-

$$\begin{array}{lcl}
 \text{eff.} & & \\
 4 \leftarrow A \rightarrow 30 \text{ min.} & & \\
 3 \leftarrow B \rightarrow 40 \text{ min} & \left. \begin{array}{l} \\ \\ \end{array} \right\} & \text{Lcm}(30, 40, 10) \\
 & & 120 \\
 12 \leftarrow A+B+C \rightarrow 10 \text{ min.} & & \\
 \downarrow \quad \downarrow \quad \downarrow & & \\
 4 \quad 3 \quad [12-(4+3)] & & \\
 5 \leftarrow C & & \\
 C \rightarrow \frac{120}{5} = \boxed{24 \text{ min.}} & \text{Ans} &
 \end{array}$$

- 8 A pipe fill a tank in 30 minutes. Due to two leakages A and B, the filled tank would be drained off in $1\frac{1}{2}$ hour and $1\frac{1}{4}$ hours respectively. How long will it take to fill the tank if the pipe, A and B are left open?

Soln:-

$$\begin{array}{lcl}
 \text{eff} & & \\
 30 \leftarrow \text{Pipe}_1 \rightarrow \frac{1}{2} \text{ hr} & & \\
 10 \leftarrow A \rightarrow \frac{3}{2} \text{ hr} & \left. \begin{array}{l} \\ \\ \end{array} \right\} & \text{Lcm}(1, 3, 5) \\
 12 \leftarrow B \rightarrow \frac{5}{4} \text{ hr} & & 15 \\
 30-10-12 \leftarrow A+B+\text{Pipe} & & \\
 \text{[8] eff} & & \\
 \text{Time} \rightarrow \frac{15}{8} & & \\
 = \boxed{1\frac{7}{8} \text{ hours}} & \text{Ans} &
 \end{array}$$

- 9] Pipe A can fill the tank in 12 hours and Pipe B can fill the tank in 8 hours. A third pipe C empties tank in 15 hours. If all pipes are opened together then after 5 hours what portion of the tank will be filled.

Soln:- m-I

$$\begin{array}{rcl} \text{eff.} & & \\ 10 \leftarrow A & \longrightarrow & 12 \text{ hr} \\ 15 \leftarrow B & \longrightarrow & 8 \text{ hr} \\ -8 \leftarrow C & \longrightarrow & 15 \text{ hr} \end{array} \left. \vphantom{\begin{array}{rcl} 10 \leftarrow A & \longrightarrow & 12 \text{ hr} \\ 15 \leftarrow B & \longrightarrow & 8 \text{ hr} \\ -8 \leftarrow C & \longrightarrow & 15 \text{ hr} \end{array}} \right\} 120$$

$$10+15-8 = A+B+C$$

$$17 = A+B+C$$

$$\text{In 5 hours} \Rightarrow 17 \times 5 = 85$$

$$\text{filled} = \frac{85}{120} \Rightarrow \boxed{\frac{17}{24}} \text{ Ans}$$

m-II Trick:

$$5 \left[\frac{1}{12} + \frac{1}{8} - \frac{1}{15} \right]$$

$$\Rightarrow 5 \left[\frac{10+15-8}{120} \right] \Rightarrow \boxed{\frac{17}{24}} \text{ Ans}$$

- 10] Three pipes x, y and z leave three different A, B and C chemicals respectively in one tank. These pipes can fill the tank in 20, 25 and 40 minutes respectively. If all the pipes are left open for 10 minutes then what will be the ratio of chemical B to the tank?

Soln:- m-I

$$\begin{array}{rcl} 10 \leftarrow A & \longrightarrow & 20 \\ 8 \leftarrow B & \longrightarrow & 25 \\ 5 \leftarrow C & \longrightarrow & 40 \end{array} \left. \vphantom{\begin{array}{rcl} 10 \leftarrow A & \longrightarrow & 20 \\ 8 \leftarrow B & \longrightarrow & 25 \\ 5 \leftarrow C & \longrightarrow & 40 \end{array}} \right\} 200$$

$$B = \frac{8 \times 10 \text{ min}}{(10+8+5) \times 10 \text{ min}}$$

$$= \boxed{\frac{8}{23}} \text{ Ans}$$

m-II

$$B = \frac{\frac{1}{25} \times 10 \text{ min}}{\left(\frac{1}{20} + \frac{1}{25} + \frac{1}{40} \right) \times 10 \text{ min}}$$

$$= \boxed{\frac{8}{23}} \text{ Ans}$$

- 11] Two pipes A and B can fill a water tank in 20 and 24 minutes respectively and a third pipe C can empty at the rate of 3 gallons per minutes. If A, B and C are opened together to fill the tank in 15 minutes, find the capacity of tank?

Soln:-

$$\begin{array}{lcl}
 6 \leftarrow A \rightarrow 20 \text{ min.} & & \\
 5 \leftarrow B \rightarrow 24 \text{ min.} & & \\
 8 \leftarrow A+B-C \rightarrow 15 \text{ min.} & \text{Total Capacity} & \\
 \downarrow \quad \downarrow & 120 & \\
 6 \quad 5 & \times 1 & \boxed{120 \text{ gallons}} \quad \underline{\underline{\text{Ans}}}
 \end{array}$$

$C = 3 \text{ Units} \xrightarrow{\times 1} 3 \text{ Gallons}$

Given

- 12] Two inlet pipes can fill a cistern in 5 and 6 hours respectively and an outlet pipe can empty 24 gallons of water per hour. All the three pipes working together can fill the empty cistern in 10 hours. What is the capacity (in gallons) of the tank?

Soln:-

$$\begin{array}{lcl}
 \text{eff} & & \\
 6 \leftarrow A \rightarrow 5 \text{ hr} & & \\
 5 \leftarrow B \rightarrow 6 \text{ hr} & & \\
 3 \leftarrow A+B-C \rightarrow 10 & \text{Total Capacity} & \\
 \downarrow \quad \downarrow \quad \downarrow & 30 & \\
 6 \quad 5 \quad -8 & &
 \end{array}$$

$C = 8 \text{ Unit} \longrightarrow 24 \text{ gallons}$

$1 \text{ Unit} \longrightarrow 3 \text{ gallons}$

Capacity = $30 \text{ Unit} = 30 \times 3 = \boxed{90 \text{ gallons}} \quad \underline{\underline{\text{Ans}}}$

13] A tank has a leak which would empty the completely filled tank in 10 hours. If the tank is full of water and a tap is opened which admits 4 litres of water per minute in the tank, the leak takes 15 hours to empty the tank. How many litres of water does the tank hold?

Soln:

$$\begin{array}{l} -3 \leftarrow -A \rightarrow 10 \text{ hrs.} \\ -2 \leftarrow -A+B \rightarrow 15 \text{ hrs} \\ \quad \downarrow \quad \downarrow \\ \quad -3 \quad \boxed{1} \end{array} \quad \rangle 30$$

$$B = 1 \text{ Unit/hr} \xrightarrow{\text{Given}} 4 \frac{\text{lit}}{\text{min}} \Rightarrow 4 \times 60 \frac{\text{lit}}{\text{hr}} = 240$$

$$\begin{aligned}\text{Capacity} &= 30 \text{ Unit} = 30 \times 240 \\ &= \boxed{7200 \text{ litres}} \quad \underline{\text{Ans}}\end{aligned}$$

14] There are two pumps to fill a tank with water. First Pumps can fill the empty tank in 8 hours, while the second in 10 hours. If both pumps are opened at the same time and kept open for 4 hours, the part of tank that will be filled up is:

Solⁿ :- $m-1$

$$\begin{array}{l} 5 \leftarrow A \rightarrow 8 \text{ hr} \\ 4 \leftarrow B \rightarrow 10 \text{ hr} \end{array} \quad \left. \vphantom{\begin{array}{l} 5 \leftarrow A \rightarrow 8 \text{ hr} \\ 4 \leftarrow B \rightarrow 10 \text{ hr} \end{array}} \right\} 40$$

$$\underline{9} \leq A+B$$

kept open for 4 hours $\Rightarrow 9 \times 4 = 36$

$$\frac{36}{40} = \boxed{\frac{9}{10}} \text{ Ans}$$

7-11

$$\left(\frac{1}{8} + \frac{1}{10}\right) \times 4 \text{ hrs}$$

$$\left(\frac{5+4}{40}\right) \times 4$$

$$\frac{9}{10} \text{ any}$$

15 A pipe can fill a tank in 24 hours. Due to a leakage in the bottom, it is filled in 36 hours. If the tank is half full, how much time will they take to empty the tank?

Soln:-

$$\begin{array}{rcl} 3 \leftarrow A \rightarrow 24 \text{ hr} & & \\ 2 \leftarrow A+B \rightarrow 36 \text{ hr.} & > & 72 \\ \downarrow \quad \downarrow & & \\ 3 \quad -1 & & \end{array}$$

$$\text{Half capacity} = \frac{72}{2} = 36 \text{ litres}$$

$$\text{Time} = \frac{36}{1} = \boxed{36 \text{ hours}} \underline{\underline{\text{Ans}}}$$

16 A tap can fill an empty tank in 12 hours and another tap can empty half the tank in 10 hours. If both the taps are opened simultaneously, how long would it take for the empty tank to be filled to half its capacity?

Soln:-

$$\begin{array}{rcl} 5 \leftarrow A \rightarrow 12 \text{ hr.} & & \\ -3 \leftarrow -B \xrightarrow{10 \times 2} 20 \text{ hr.} & > & 60 \end{array}$$

$$5-3 = A+B$$

$$2 = A+B$$

$$A+B \rightarrow \frac{\frac{60}{2}}{2} = \boxed{15 \text{ hours}} \underline{\underline{\text{Ans}}}$$

17] A tap can fill a tank in 3 hours and a leak in the bottom can empty it in 5 hours. If tap and leakage (half closed) are opened together then find the time taken to fill the tank.

Soln:

eff

$S \leftarrow A \rightarrow 3 \text{ km}$
 $3 \leftarrow B \rightarrow 5 \text{ hrs.}$ $\rangle 15$

$$\text{efficiency} = 5 - \frac{3}{2} (\text{half closed}) = \frac{7}{2}$$

$$\text{Time} = \frac{15}{7/2} = \frac{30}{7} \Rightarrow \boxed{4\frac{2}{7} \text{ hour}} \underline{\underline{\text{Ans}}}$$

18. A pump can fill a tank with water in 1 hour. Because of a leak, it took $1\frac{1}{3}$ hours to fill the tank. In how many hours can the leak alone drain all the water of the tank when it is full?

Soln:-

$$\begin{array}{l} 4 \leftarrow A \rightarrow 1 \text{ hr.} \\ 3 \leftarrow A-B \rightarrow \frac{4}{3} \text{ hr.} \\ \quad \downarrow \quad \downarrow \\ \quad 4 \quad -1 \end{array} \quad > \quad \begin{array}{c} \text{Capacity} \\ 4 \end{array}$$

eff. $B = -1$

$$\begin{aligned}\text{Time} &= \frac{4}{1} \\ &= \boxed{4 \text{ hours}} \text{ Ans}\end{aligned}$$

19] A tap can fill a tank in $5\frac{1}{2}$ hours. Because of a leak, it took $8\frac{1}{4}$ hours to fill the tank. In how much time (in hours) will the leak alone empty 30% of the tank?

Soln :-

$$\begin{array}{l} G \leftarrow A \rightarrow \frac{11}{2} \text{ hr} \\ 4 \leftarrow A-B \rightarrow \frac{33}{4} \\ \quad \downarrow \quad \downarrow \\ \quad 6 \quad -2 \end{array} \quad \rangle \quad 33$$

$$B = -2$$

$$\text{Time} = \frac{33 \times 30}{100}$$

11 $\frac{99}{20}$ hour Ans

20. Two pipes A and B can fill cistern in $12\frac{1}{2}$ hours and 25 hours, respectively. The pipes were opened simultaneously, and it was found that, due to leakage in the bottom, it took one hour 40 minutes more to fill the cistern. If the cistern is full, in how much time (in hours) will the leak alone empty 70% of the cistern?

Soln:-

$2 \leftarrow A \rightarrow \frac{25}{2} \text{ hour}$
 $1 \leftarrow B \rightarrow 25 \text{ hour}$

$$A+B \Rightarrow \frac{25}{(2+1)} = 8 \text{ hr } 20 \text{ min},$$

$$A+B-C \rightarrow 8\text{hr } 20\text{min.} + 2\text{hr. } 40\text{min.}$$

$$\Rightarrow 10\text{hr.}$$

$$\begin{array}{l} 8 \leftarrow A \rightarrow \frac{25}{2} \\ 4 \leftarrow B \rightarrow 25 \\ 10 \leftarrow A+B-C \rightarrow 10 \end{array} \quad \left. \vphantom{\begin{array}{l} 8 \leftarrow A \rightarrow \frac{25}{2} \\ 4 \leftarrow B \rightarrow 25 \\ 10 \leftarrow A+B-C \rightarrow 10 \end{array}} \right\} 100$$

$$\text{eff. } C = 8 + 4 - 10 = \boxed{2}$$

$$\text{Time} = \frac{100 \times 70\%}{2}$$
$$= \boxed{35 \text{ hrs}}$$

Ans

- [21] A tap takes 36 hours extra to fill a tank due to a leakage equivalent to half of its inflow. The inflow can fill the tank in how many hours?

Soln:-

	A	B
efficiency $\rightarrow$	2	: 1
Time $\rightarrow$	1	: 2

$\swarrow \times 36$
 $\boxed{36 \text{ hrs}}$

$\ominus$
 1 Unit $\rightarrow 36 \text{ hr.}$
Ans

- [22] The volume of a tank is 72 cubic metres. Water is poured into it at the rate of 60 litres per minute. How much time will it take to fill the tank?

Soln:-

$$\text{Time} = \frac{72000 \text{ lit}}{60 \text{ lit/min}} \quad [1 \text{ m}^3 = 1000 \text{ lit}]$$

$$= \frac{1200 \text{ min.}}{60} = \boxed{20 \text{ hours}} \quad \underline{\text{Ans}}$$

MathswithGajanand

- [23] $\frac{3}{4}$ part of a tank is filled with oil. After taking out 60 litres of oil, the tank is $\frac{2}{3}$ part full. What is the capacity (in litres) of the tank?

Soln:- mf Basic let Capacity = x

$$\frac{2}{3}x = \frac{3}{4}x - 60$$

$$\frac{3}{4}x - \frac{2}{3}x = 60$$

$$\frac{x}{12} = 60$$

$$x = \boxed{720 \text{ lit}}$$

let Capacity

$$\text{LCM}\left(\frac{3}{4}, \frac{2}{3}\right) = 12$$

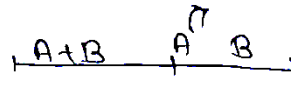
$$\begin{array}{ccc} & \text{LCM} & \\ \swarrow \times \frac{3}{4} & & \searrow \times \frac{2}{3} \\ 9 & \ominus & 8 \\ & \text{1 Unit} \rightarrow 60 \text{ lit} & \end{array}$$

$$\text{Capacity} = 12 \text{ Unit} = 12 \times 60$$

$$\boxed{720 \text{ litres}} \quad \underline{\text{Ans}}$$

Ex-II

$$\begin{array}{l} 3 \leftarrow A \rightarrow 20 \\ 2 \leftarrow B \rightarrow 30 \end{array} \left. \vphantom{\begin{array}{l} 3 \leftarrow A \rightarrow 20 \\ 2 \leftarrow B \rightarrow 30 \end{array}} \right\} 60$$



Here B is working for 18 minutes starting to last.

$$\begin{aligned} \text{So work done by B} &= 2 \times 18 \\ &= 36 \end{aligned}$$

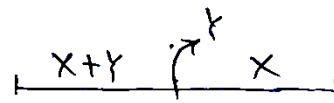
$$\begin{aligned} \text{Remaining work} &= 60 - 36 \\ &= 24 \end{aligned}$$

This 24 Unit must be filled by Pipe A in beginning

$$\begin{aligned} \text{Time A} &= \frac{24}{3} \\ &= \boxed{8 \text{ min.}} \text{ Ans} \end{aligned}$$

26 Two pipes X and Y can fill a cistern in 24 minutes and 32 minutes respectively. If both the pipes are opened together, then after how much time (in minutes) should Y be closed so that the tank is full in 18 minutes?

$$\begin{array}{l} \text{Soln :-} \\ \begin{array}{l} 4 \leftarrow X \rightarrow 24 \\ 3 \leftarrow Y \rightarrow 32 \end{array} \end{array} \left. \vphantom{\begin{array}{l} 4 \leftarrow X \rightarrow 24 \\ 3 \leftarrow Y \rightarrow 32 \end{array}} \right\} \begin{array}{l} \text{eff} \\ \text{Capacity} \end{array} 96$$



$$\begin{aligned} \text{Pipe X fills in 18 min} &= 4 \times 18 \\ &= 72 \end{aligned}$$

$$\begin{aligned} \text{Remaining} &= 96 - 72 \\ &= 24 \end{aligned}$$

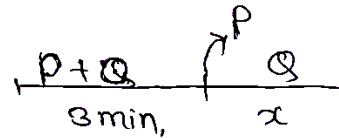
$$\begin{aligned} \text{Y fills} &= \frac{24}{3} \\ &= \boxed{8 \text{ min.}} \text{ Ans} \end{aligned}$$

- 24** Two pipes, P and Q can fill a cistern in 12 and 15 minutes respectively. Both are opened together, but at the end of 3 minutes, P is turned off. In how many more minutes will Q fill the cistern?

Soln:-

eff.

$$\begin{array}{l} 5 \leftarrow P \rightarrow 12 \\ 4 \leftarrow Q \rightarrow 15 \end{array} > 60$$



$$(5+4) \times 3 \text{ min} + 4x = 60$$

$$4x = 60 - 27$$

$$x = \frac{33}{4}$$

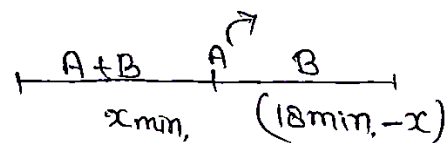
$$= \boxed{8 \frac{1}{4} \text{ min}} \quad \underline{\underline{\text{Ans}}}$$

- 25** A tank can be filled by two pipes in 20 minutes and 30 minutes respectively. When the tank was empty, the two pipes were opened. After some time, the first pipe was stopped and the tank was filled in 18 minutes. After how much time of the start the first stopped?

Soln:-

eff

$$\begin{array}{l} 3 \leftarrow A \rightarrow 20 \\ 2 \leftarrow B \rightarrow 30 \end{array} > \begin{array}{l} \text{Capacity} \\ 60 \end{array}$$



$$(3+2)x + 2 \times (18-x) = 60$$

$$5x + 36 - 2x = 60$$

$$3x = \frac{24}{1}$$

$$x = \boxed{8 \text{ minutes}}$$

Q7 A water tank can be filled by a tap in 30 minutes and another tap can fill it in 60 minutes. If both the taps are kept open for 5 minutes and then the first tap is closed, how long will it take for the tank to be filled?

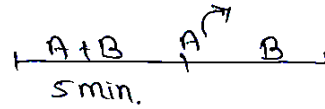
Soln :- eff.

$$\begin{array}{rcl} 2 & \leftarrow A \rightarrow & 30 \\ 1 & \leftarrow B \rightarrow & 60 \end{array} > 60$$

$$-15$$

$$\text{Remaining} = 45$$

$$\text{Time} = \frac{45}{1} = \boxed{45 \text{ min}} \text{ Ans}$$



$$(2+1) \times 5 \text{ min} = 15 \text{ Unit}$$

m-II $(2+1) \times 5 \text{ min} + 1 \times x = 60$ [x is B's time]

$$15 + x = 60$$

$$x = \boxed{45 \text{ min}} \text{ Ans}$$

Q8 A tank can be filled with water by two pipes, A and B together in 36 minutes. If the pipe B was stopped after 30 minutes, the tank is filled in 40 minutes. The Pipe B can alone fill the tank in?

Soln :-

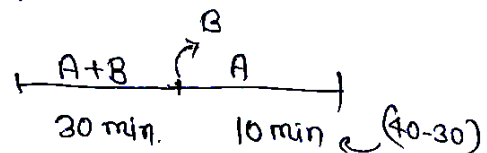
$$\begin{array}{l} \text{Total work} \\ (A+B) \times 36 = 30 \times (A+B) + 10A \end{array}$$

$$(A+B) \times 6 = 10A$$

$$6B = 4A$$

$$\text{efficiency } \boxed{A : B = 3 : 2}$$

$$\begin{aligned} \text{Total work} &= (A+B) \times 36 \\ &= 5 \times 36 \\ &= 180 \end{aligned}$$

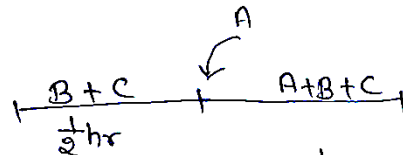


$$\begin{aligned} B &\rightarrow \frac{180}{2 \text{ eff.}} \\ &= \boxed{90 \text{ min.}} \text{ Ans}$$

- 29] Three pipes A, B and C can fill a tank in 6 hours, 9 hours and 12 hours respectively. B and C are opened for half an hour, then A is also opened. The time taken by the three pipes together to fill the remaining part of the tank is:

Soln:-

$$\begin{array}{l} \text{eff.} \\ 6 \leftarrow A \rightarrow 6 \\ 9 \leftarrow B \rightarrow 9 \\ 12 \leftarrow C \rightarrow 12 \end{array} \left. \vphantom{\begin{array}{l} 6 \\ 9 \\ 12 \end{array}} \right\} 36$$



$$\begin{aligned} (4+3) \times \frac{1}{2} \\ = \frac{7}{2} \text{ Unit} \end{aligned}$$

$$\frac{\frac{65}{2}}{(6+4+3)}$$

$$\begin{aligned} \text{Remaining} &= 36 - \frac{7}{2} \\ &= \frac{65}{2} \text{ Unit} \end{aligned}$$

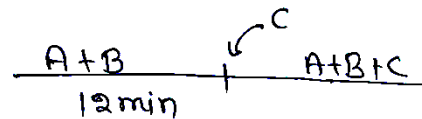
$$\frac{65}{2 \times 13}$$

$$= \boxed{2 \frac{1}{2} \text{ hr}} \text{ Ans}$$

- 30] Two pipes A and B can fill a tank with water in 30 minutes and 45 minutes respectively. The third pipe C can empty the tank in 36 minutes. First A and B are opened. After 12 minutes C is opened. Total time in which the tank will be filled up is:

Soln:-

$$\begin{array}{l} 6 \leftarrow A \rightarrow 30 \\ 45 \leftarrow B \rightarrow 45 \\ -5 \leftarrow C \rightarrow 36 \end{array} \left. \vphantom{\begin{array}{l} 6 \\ 45 \\ -5 \end{array}} \right\} \begin{array}{l} \text{Capacity} \\ 180 \end{array}$$



$$\text{In 12 min} \Rightarrow 12 \times 10 = 120$$

$$\text{Remaining} = 180 - 120 = 60$$

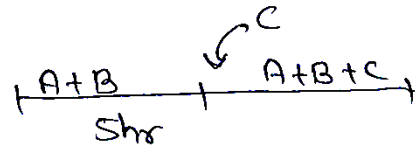
$$\text{Time} = \frac{60}{6+4-5} = 12 \text{ min.}$$

$$\text{Total time} = \boxed{24 \text{ min}}$$

[31] Two pipes A and B can fill an empty tank in 10 hours and 15 hours respectively. Pipe C alone can empty the completely filled tank in 12 hours. First both pipes A and B are opened and after 5 hours pipe C is also opened. What is the total time (in hours) in which the tank will be filled?

Soln:-

$$\begin{array}{l} 6 \leftarrow A \rightarrow 10 \\ 4 \leftarrow B \rightarrow 15 \\ -5 \leftarrow C \rightarrow 12 \end{array} \left. \vphantom{\begin{array}{l} 6 \\ 4 \\ -5 \end{array}} \right\} 60$$



$$\text{in 5 hr} \rightarrow 5 \times 10 = 50$$

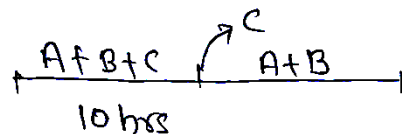
$$\begin{aligned} \text{Remaining} &\rightarrow 60 - 50 = \frac{10}{(6+4-5)} \\ &= \boxed{2 \text{ hr}} \end{aligned}$$

$$\text{Total time} = \boxed{7 \text{ hr}} \text{ Ans}$$

[32] Pipe A and B can fill a tank in 16 hours and 24 hours, respectively, whereas pipe C can empty the full tank in 40 hours. All three pipes are opened together, but pipe C is closed after 10 hours. After how many hours will the

Soln:-

$$\begin{array}{l} 15 \leftarrow A \rightarrow 16 \\ 10 \leftarrow B \rightarrow 24 \\ -6 \leftarrow C \rightarrow 40 \end{array} \left. \vphantom{\begin{array}{l} 15 \\ 10 \\ -6 \end{array}} \right\} 240$$



$$\begin{aligned} \text{In 10 hrs} &\rightarrow (15+10-6) \times 10 \\ &= 190 \end{aligned}$$

$$\text{Remaining} = 240 - 190 = 50$$

$$= \frac{50}{(15+10)} = \boxed{2 \text{ hr}} \text{ Ans}$$

- 33] Pipes A and B are filling pipes while pipe C is an emptying pipe. A and B can fill a tank in 72 and 90 min, respectively. When all the three pipes are opened together, the tank gets filled in 2 hours. A and B are opened together for 12 minutes, then closed and C is opened. The tank will be empty after!

Soln:-

$$\begin{array}{lcl}
 5 \leftarrow A \rightarrow & 72 \text{ min} & \\
 4 \leftarrow B \rightarrow & 90 \text{ min} & \\
 3 \leftarrow A+B+C \rightarrow & 120 \text{ min} & \\
 \downarrow \downarrow \downarrow & & \\
 5 & 4 & -6
 \end{array}
 \begin{array}{l}
 \text{Total} \\
 \text{Capacity} \\
 360
 \end{array}$$

$$\begin{array}{c}
 \text{A+B} \quad \text{C} \\
 \hline
 12 \text{ min}
 \end{array}$$

In 12 min $\rightarrow 12 \times 9 = 108$

Remaining $\Rightarrow \frac{108}{6} = 18 \text{ min}$ Ans

- 34] Three pipes A, B and C can fill a cistern in 6 hours. After working together for 2 hours, C is closed and A and B fill it in 7 hours more. The time taken by C alone to fill the cistern is?

Soln:- let total Capacity = 42 Unit

eff. of $(A+B+C) = \frac{42}{6} = 7 \text{ Unit/hr}$

$$\begin{array}{c}
 \text{A+B+C} \quad \text{C} \\
 \hline
 2 \text{ hr} \quad 7 \text{ hr}
 \end{array}$$

In 2 hr $\rightarrow 14 \text{ Unit}$

Remaining $= 42 - 14 = 28 \leftarrow \text{In 7 hr}$

efficiency of $A+B = \frac{28}{7} = 4 \text{ Unit/hr}$

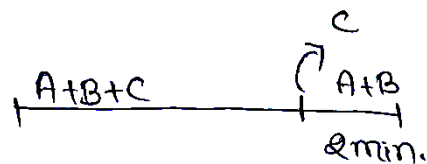
efficiency of $C = (A+B+C) - (A+B)$
 $= 7 - 4$
 $= 3$

$C \rightarrow \frac{42}{3} = 14 \text{ hrs}$ Ans

35 Two pipes A and B can fill a tank in 12 minutes and 24 minutes, respectively, while a third pipe C can empty the full tank in 32 minutes. All the three pipes are opened simultaneously. However, pipe C is closed 2 minutes before the tank is filled. In how much time (in minutes) will the tank be full?

Soln:-

$$\begin{array}{l} 8 \leftarrow A \rightarrow 12 \\ 4 \leftarrow B \rightarrow 24 \\ 3 \leftarrow C \rightarrow 32 \end{array} \rightarrow 96$$



in 2 min $\rightarrow 2 \times (8+4) = 24$

Remaining = $96 - 24 = 72$

$= \frac{72}{(8+4-3)} = \boxed{8 \text{ min}}$

Total time = $8 + 2 \text{ min} = \boxed{10 \text{ min}}$ Ans

MathswithGajanand

m-II

If C work 2 min more

then total work = $96 - 3 \times 2$
 $= 90$

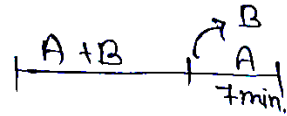
Time = $\frac{90}{(8+4-3)}$

$= \boxed{10 \text{ min}}$ Ans

- [36]** Two fill pipes A and B can fill cistern in 56 and 63 min. respectively. Both fill pipes are opened together. But 7 minutes before the cistern will fill B is closed. In how much time will the cistern take to fill?

Soln:-

$$\begin{array}{l} 9 \leftarrow A \rightarrow 56 \\ 8 \leftarrow B \rightarrow 63 \end{array} \quad \left. \begin{array}{l} \\ \end{array} \right\} 7 \times 9 \times 8 = 504$$



$$\Rightarrow 504 + 7 \times 8 \quad \leftarrow \text{Extra work by B}$$

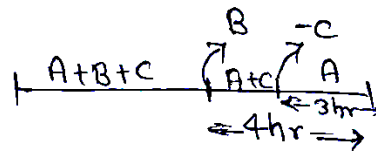
$$\Rightarrow \frac{560}{(8+9)} = \boxed{32 \frac{16}{17} \text{ min}}$$

Ans

- [37]** Two pipes could fill an empty tank in 20 hrs, 28 hrs. respectively while a third one can drain the filled tank in 35 hours. If the outlet pipe was closed 3 hours before the tank got filled and second pipe which can filled cistern in 28 hrs also closed 4 hrs. before the cistern is filled, how many hours since the beginning did it take to fill the tank?

Soln:-

$$\begin{array}{l} 7 \leftarrow A \rightarrow 20 \\ 5 \leftarrow B \rightarrow 28 \\ -4 \leftarrow C \rightarrow 35 \end{array} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} 140$$



$$140 + 4 \times 5 - \underbrace{3 \times 4}_{\text{C's work}}$$

$\downarrow$
B's extra work

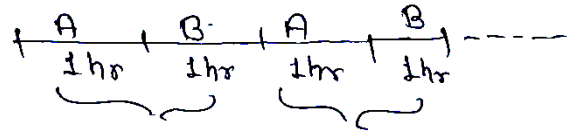
$$= 140 + 20 - 12 = 148$$

$$\text{Time} = \frac{148}{(7+5-4)} = \boxed{18.5 \text{ hr}} \quad \underline{\underline{\text{Ans}}}$$

38 Pipe A can fill a tank in 4 hours and pipe B can fill it in 6 hours. If they are opened on alternate hours and if pipe A is opened first then in how many hours, the tank shall be full?

Soln:-

$$\begin{array}{l} 3 \leftarrow A \rightarrow 4 \text{ hr} \\ 2 \leftarrow B \rightarrow 6 \text{ hr} \end{array} \rangle 12$$



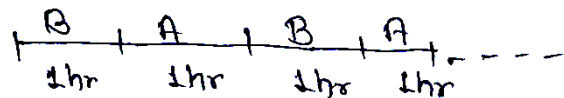
$$\begin{array}{l} \text{In 2 hrs} \rightarrow 5 \text{ Unit} \quad \leftarrow A \text{ } B \\ \times 2 \qquad \qquad \times 2 \end{array}$$

$$\begin{array}{rcl} 4 \text{ hrs} & \rightarrow & 10 \text{ Unit} \\ A \rightarrow + \frac{2}{3} \text{ hr} & & + 2 \text{ Unit} \\ \hline & & 12 \text{ Unit} \\ \boxed{4 \frac{2}{3}} & \text{Ans} & \end{array}$$

39 Pipe A can fill a tank in 12 hours and Pipe B can fill the tank in 18 hours. If both the pipes are opened on alternate hours and if pipe B is opened first, then in how much time (in hours) the tank will be full?

Soln:-

$$\begin{array}{l} 3 \leftarrow A \rightarrow 12 \text{ hr} \\ 2 \leftarrow B \rightarrow 18 \text{ hr} \end{array} \rangle 36$$



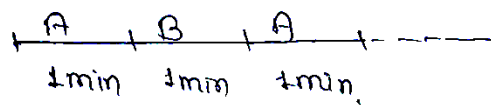
$$\begin{array}{rcl} \text{In 2 hrs} & \rightarrow & 5 \text{ Unit} \\ \times 7 & & \times 7 \\ \hline 14 \text{ hr} & \rightarrow & 35 \text{ Unit} \\ B \rightarrow \frac{1}{2} \text{ hr} & & + 1 \\ \hline \boxed{14 \frac{1}{2} \text{ hr}} & \text{Ans} & \end{array}$$

40] A cistern is provided with two pipes A and B.

A can fill it in 20 minutes and B can empty it in 30 minutes. If A and B be kept open alternatively for one minutes each, how soon will the cistern be filled?

Soln:-

$$\begin{array}{l} 3 \leftarrow A \rightarrow 20 \text{ min} \\ 2 \leftarrow B \rightarrow 30 \text{ min} \end{array} > 60$$

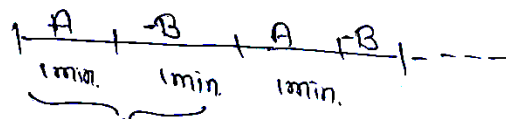


$$\begin{array}{rcl} \text{In 2 min.} & \rightarrow & (3-2) \text{ Unit} \\ & & 1 \text{ Unit} \\ \hline & & \times 57 \\ \hline 114 \text{ min} & \rightarrow & 57 \text{ Unit} \\ A \rightarrow & 1 \text{ min} & \rightarrow +3 \\ \hline & & 60 \\ \hline \boxed{115 \text{ min.}} & \text{Ans} & \end{array}$$

41] A tank is fitted with two taps. The first tap can fill the tank completely in 45 minutes and the second tap can empty the full tank in one hour. If both the taps are opened alternately for one minutes, then in how many hours the empty tank will be filled Completely

Soln:- eff

$$\begin{array}{l} 4 \leftarrow A \rightarrow 45 \text{ min} \\ 3 \leftarrow B \rightarrow 60 \text{ min} \end{array} > 180$$

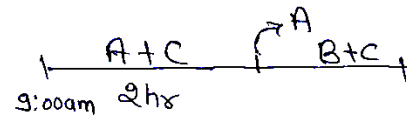


$$\begin{array}{rcl} 2 \text{ min.} & \rightarrow & (4-3) \text{ Unit} \\ & & 1 \text{ Unit} \\ \hline & & \times 176 \\ \hline 352 \text{ min} & \rightarrow & 176 \text{ Unit} \\ A \rightarrow & 1 \text{ min} & \rightarrow +4 \text{ Unit} \\ \hline & & 180 \\ \hline \boxed{353 \text{ min}} & & \\ = & \underline{\underline{5 \text{ hr. } 53 \text{ min}}} & \text{Ans} \end{array}$$

- 42] Pipes A and B can fill a tank in one hour and two hours respectively while pipe C can empty the filled up tank in one hour and 15 minutes. A and C are turned on together at 9 a.m. After 2 hours. Only A is closed and B is turned on. When will the tank be emptied?

Soln:-

$$\begin{array}{l} 10 \leftarrow A \rightarrow 1\text{hr} \\ 5 \leftarrow B \rightarrow 2\text{hr} \\ -8 \leftarrow C \rightarrow \frac{5}{4}\text{hr} \end{array} \left. \vphantom{\begin{array}{l} 10 \\ 5 \\ -8 \end{array}} \right\} 10$$



$$(A+C) \rightarrow \text{in } 2\text{hr} \Rightarrow (10-8) \times 2 = 4$$

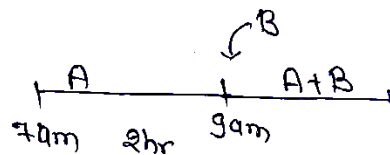
$$(B+C) \Rightarrow \frac{4}{(5-8)} = \frac{4}{3} = 1\text{hr } 20\text{min}$$

$$9\text{am} + 2\text{hr} + 1\text{hr } 20\text{min} = \boxed{12:20\text{pm}} \quad \underline{\text{Ans}}$$

- 43] Pipe A can fill the tank in 8 hours and pipe B can fill it in 12 hours. If pipe A is opened at 7:00 am and pipe B is opened at 9:00 am, then at what time will the tank be full?

Soln:-

$$\begin{array}{l} 3 \leftarrow A \rightarrow 8\text{hr} \\ 2 \leftarrow B \rightarrow 12\text{hr} \end{array} \left. \vphantom{\begin{array}{l} 3 \\ 2 \end{array}} \right\} 24$$



m-I

$$A \rightarrow \text{In } 2\text{hr} \Rightarrow 3 \times 2 = 6$$

$$\begin{aligned} \text{Remaining} &= 24 - 6 = \frac{18}{5} \\ &= 3\text{hr } 36\text{min} \end{aligned}$$

$$9\text{am} + 3\text{hr } 36\text{min} = \boxed{12:36\text{pm}} \quad \underline{\text{Ans}}$$

m-II

$$3 \times 2 + (3+2) \times x = 24$$

$$6 + 5x = 24$$

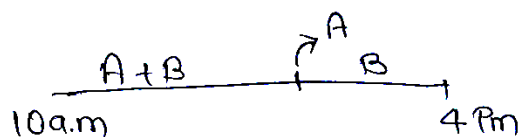
$$x = \frac{18}{5} = 3\text{hr } 36\text{min}$$

$$9\text{am} + 3\text{hr } 36\text{min} = \boxed{12:36\text{pm}} \quad \underline{\text{Ans}}$$

[44] Two taps A and B can fill a tank in 10 hours and 12 hours respectively. If the two taps are opened at 10 a.m., then at what time (in p.m.) should the tap A be closed to completely fill the tank at exactly 4 p.m?

Soln:-

$$\begin{array}{l} 6 \leftarrow A \rightarrow 10 \text{ hr} \\ 5 \leftarrow B \rightarrow 12 \text{ hr} \end{array} \left. \vphantom{\begin{array}{l} 6 \leftarrow A \rightarrow 10 \text{ hr} \\ 5 \leftarrow B \rightarrow 12 \text{ hr} \end{array}} \right\} 60$$



[m-I] Work done by B from (10 a.m. - 4 p.m.)

$$\begin{aligned} &= 5 \times 6 \text{ hr} \\ &= 30 \end{aligned}$$

$$\begin{aligned} \text{Remaining} &= 60 - 30 \\ \text{work by A} &= \frac{30}{6} = 5 \text{ hr} \end{aligned}$$

$$10 \text{ a.m.} + 5 \text{ hr} = \boxed{3:00 \text{ p.m.}}$$

[m-II] Basic

$$10 \text{ a.m.} - 4 \text{ p.m.} = 6 \text{ hr}$$

$$(6+5)x \text{ hr} + 5 \times (6-x) = 60$$

$$11x + 30 - 5x = 60$$

$$x = 5 \text{ hr}$$

$$10 \text{ a.m.} + 5 \text{ hr} = \boxed{3 \text{ p.m.}}$$

Ans

[45] Pipes P and Q can fill a tank in 10 hours and 12 hours respectively and C can empty it in 6 hours. If all the three are opened at 7 a.m., at what time will one-fourth of the tank be filled?

Soln:-

$$\begin{array}{l} 6 \leftarrow P \rightarrow 10 \\ 5 \leftarrow Q \rightarrow 12 \\ 10 \leftarrow C \rightarrow 6 \end{array} \left. \vphantom{\begin{array}{l} 6 \leftarrow P \rightarrow 10 \\ 5 \leftarrow Q \rightarrow 12 \\ 10 \leftarrow C \rightarrow 6 \end{array}} \right\} 60$$

$$\begin{aligned} \frac{1}{4}^{\text{th}} \text{ of total work} \\ &= 60 \times \frac{1}{4} = 15 \end{aligned}$$

$$\text{Time} = \frac{15}{(6+5-10)} = 15 \text{ hr}$$

$$7 \text{ a.m.} + 15 \text{ hr} \Rightarrow \boxed{10 \text{ p.m.}} \quad \underline{\underline{\text{Ans}}}$$

46 Pipes A, B and C can fill a tank in 20, 30 and 60 hours, respectively. Pipes A, B and C are opened at 7 a.m., 8 a.m. and 9 a.m. respectively, on the same day. When will the tank be full?

Soln:-

eff.

$$\begin{array}{l} 3 \leftarrow A \rightarrow 20 \\ 2 \leftarrow B \rightarrow 30 \\ 1 \leftarrow C \rightarrow 60 \end{array} \left. \vphantom{\begin{array}{l} 3 \leftarrow A \rightarrow 20 \\ 2 \leftarrow B \rightarrow 30 \\ 1 \leftarrow C \rightarrow 60 \end{array}} \right\} 60$$

Till 9 a.m. $\Rightarrow A = 3 \times 2 \text{ hr} = 6$

$$B = 2 \times 1 \text{ hr} = 2$$

$$8$$

After 9 a.m. $\Rightarrow 60 - 8 = \frac{52}{(3+2+1)} = 8 \text{ hr } 40 \text{ min}$

9 a.m. + 8 hr 40 min = **5:40 PM** Ans

47 A water reservoir has two inlets and one outlet. Through the inlet it can be filled in 3 hours and 3 hours 45 minutes respectively. It can be emptied completely in 1 hour by the outlet. If the two inlets are opened at 01:00 pm and 02:00 pm respectively and the outlet at 3:00 pm then it will be emptied at:

Soln:-

$$\begin{array}{l} 5 \leftarrow A \rightarrow 3 \text{ hr} \\ 4 \leftarrow B \rightarrow \frac{15}{4} \text{ hr} \\ 15 \leftarrow C \rightarrow 1 \end{array} \left. \vphantom{\begin{array}{l} 5 \leftarrow A \rightarrow 3 \text{ hr} \\ 4 \leftarrow B \rightarrow \frac{15}{4} \text{ hr} \\ 15 \leftarrow C \rightarrow 1 \end{array}} \right\} 15$$

Till 3 pm

A fills $\rightarrow 5 \times 2 = 10 \text{ Unit}$

B fills $\rightarrow 4 \times 1 = 4 \text{ Unit}$

$$14 \text{ Unit}$$

eff. of $A+B-C = 5+4-15 = -6$

Tank will be empty in $= \frac{14}{6}$

$= 2 \text{ hr } 20 \text{ min}$

3 pm + 2 hr 20 min = **5:20 pm**

48 Pipes A, B and C can fill a tank in 15, 30 and 40 hours, respectively. Pipe A, B and C are opened at 6 a.m., 8 a.m. and 10 a.m., respectively, on the same day. When will the tank be full?

Soln:-

$$\begin{array}{l} 8 \leftarrow A \rightarrow 15 \\ 4 \leftarrow B \rightarrow 30 \\ 3 \leftarrow C \rightarrow 40 \end{array} \left. \vphantom{\begin{array}{l} 8 \\ 4 \\ 3 \end{array}} \right\} 120$$

Till 10am

$$A \text{ fills} \Rightarrow 8 \times 4 = 32 \text{ Unit}$$

$$B \text{ fills} \Rightarrow 4 \times 2 = 8 \text{ Unit}$$

$$\underline{40 \text{ Unit}}$$

$$\text{Remaining} = 120 - 40 = 80$$

$$A+B+C \rightarrow \frac{80}{(8+4+3)} = 5 \text{ hr } 20 \text{ min}$$

$$10 \text{ am} + 5 \text{ hr } 20 \text{ min} = \boxed{3:20 \text{ pm}} \text{ Ans}$$

49 When operated separately, pipe A takes 5 hours less than pipe B to fill a cistern, and when operated together, the cistern gets filled in 6 hours. In how much time (in hours) will pipe A fill the cistern, if operated separately?

Soln:- (a) 10

(b) 18

(c) 9

(d) 15

m-I Basic

$$A \rightarrow x$$

$$B \rightarrow x+5$$

$$A+B \rightarrow 6 \text{ hr}$$

$$6 = \frac{x(x+5)}{2x+5}$$

$$x^2 - 7x - 30 = 0$$

$$\boxed{x=10}$$

$$A = \boxed{10 \text{ hr}} \text{ Ans}$$

m-II by options

$$\text{let } A = 10 \text{ hours}$$

$$B \rightarrow 15 \text{ hr}$$

$$A+B \rightarrow \frac{10 \times 15}{10+15} = 6 \text{ hr}$$

satisfy ✓

50 12 pumps working 6 hours a day can empty a completely filled reservoir in 15 days. How many such pumps working 9 hours a day will empty the same reservoir in 12 days.

Soln:-

$$\frac{m_1 D_1 h_1}{w_1} = \frac{m_2 D_2 h_2}{w_2}$$

$$12 \times 6 \times 15 = x \times 9 \times 12$$

$$x = \boxed{10 \text{ pumps}} \text{ } \underline{\underline{\text{Ans}}}$$

51 Having the same capacity 9 taps fill up a water tank in 20 minutes. How many taps of the same capacity are required to fill up the same water tank in 15 minutes?

Soln:-

$$9 \times 20 = T \times 15$$

$$T = \boxed{12 \text{ Taps}} \text{ } \underline{\underline{\text{Ans}}}$$

52 In what time would a cistern be filled by three whose diameters are 1cm, $1\frac{1}{3}$ cm, 2cm running together, when the largest alone fill it in 122 minutes, the amount of water flowing in by each pipe being proportional to the square of its diameter?

Soln:-

$$\text{Diameter} \rightarrow 1 : \frac{4}{3} : 2$$

$$\text{Efficiency} \rightarrow (1)^2 : \left(\frac{4}{3}\right)^2 : 2^2$$

$$\Rightarrow 1 : \frac{16}{9} : 4$$

$$= 9 : 16 : 36$$

$$\text{Capacity} = 36 \times 122$$

they fill combine

$$= \frac{36 \times 122}{61}$$

$$= \boxed{72 \text{ min}} \text{ } \underline{\underline{\text{Ans}}}$$

[53] A pipe can empty a tank in 40 minutes. A second pipe with diameter twice as much as that of the first is also attached with the tank to empty it. The two pipes together can empty the tank in:

Soln:-

	A	B
diameter	D	2D
	1	2
efficiency	$(1)^2$	$(2)^2$
	1	4
Time	40	10
	$\downarrow \times 10$	$\downarrow \times 10$
	40 min	10 min
efficiency	1	4
	40 (Total Capacity)	
A+B	$\rightarrow \frac{40}{4+1} = \boxed{8 \text{ minutes}} \text{ ans}$	

[54] A cylindrical cistern of diameter 25 cm is full of water. If 11 litres of water is drawn off, the water level in the cistern will drop by?

Soln:-

$$\text{Volume} = \pi r^2 h$$

$$[1 \text{ litre} = 1000 \text{ cm}^3]$$

$$\frac{22}{7} \times \frac{25}{2} \times \frac{25}{2} \times h = 11000 \text{ cm}^3$$

$$h = \frac{112}{5}$$

$$h = \boxed{22 \frac{4}{5} \text{ cm}} \text{ ans}$$

55 Which of these pipes will empty the pool fastest?

- (a) One pipe of diameter 60m
- (b) Two pipe of diameter 30cm
- (c) Three pipes of diameter 20cm
- (d) None of these

Soln:- diameter $\rightarrow 60 : 30 : 20$
 $6 : 3 : 2$

efficiency $\rightarrow (6)^2 : (3)^2 : (2)^2$

$$= 36 : 9 : 4$$

$$\text{No. of Pipe} \rightarrow \frac{x_1}{36} : \frac{x_2}{18} : \frac{x_3}{12}$$

So pipe 1 with diameter 60m is fastest

56 One fill pipe A takes $21\frac{1}{3}$ minutes more to fill the cistern than two fill pipes A and B opened together to fill it. Second fill pipe B takes 27 minutes more to fill the cistern than two fill pipes A and B opened together to fill it. When will the cistern be full if both the pipes are opened simultaneously?

Soln:-

$$A + B \rightarrow x$$

$$A \rightarrow x + \frac{64}{3}$$

$$B \rightarrow x + 27$$

$$x = \sqrt{\frac{64}{3} \times 27}$$

$$= \boxed{24 \text{ min}} \quad \underline{\underline{\text{Ans}}}$$